1 Equations in ρ coordinates

$$0 = \frac{1}{V'}\frac{\partial}{\partial\rho}\left[V'\langle|\nabla\rho|^2\rangle\frac{\partial}{\partial\rho}\Phi\right] + \frac{1}{\epsilon_0}\left(-en_\mathrm{e} + Z_\mathrm{i}en_\mathrm{i} + Z_\mathrm{z}en_\mathrm{z} + Z_\mathrm{b}en_\mathrm{b} + Z_\mathrm{b}egn_\mathrm{b}^\mathrm{rp}\right) \tag{1.1}$$

$$\begin{aligned}\frac{1}{c^2}\frac{\partial}{\partial t}\dot{\psi}_t = {} & V'\left\langle\frac{|\nabla\rho|^2}{R^2}\right\rangle\frac{\partial}{\partial\rho}\left[\frac{1}{V'\langle R^{-2}\rangle}\frac{\partial}{\partial\rho}\psi_t\right] \\ & + \frac{\mu_0}{\langle B^\theta\rangle}\left[-en_\mathrm{e}\left(\langle Bu_{\mathrm{e}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{e}\zeta}}{R}\right\rangle\right) + Z_\mathrm{i}en_\mathrm{i}\left(\langle Bu_{\mathrm{i}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{i}\zeta}}{R}\right\rangle\right)\right. \\ & \left. + Z_\mathrm{z}en_\mathrm{z}\left(\langle Bu_{\mathrm{z}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle\right) + Z_\mathrm{b}en_\mathrm{b}\left(\langle Bu_{\mathrm{b}\parallel}\rangle - I\left\langle\frac{u_{\mathrm{b}\zeta}}{R}\right\rangle\right)\right]\end{aligned} \tag{1.2}$$

$$\begin{aligned}\frac{1}{c^2}\frac{\partial}{\partial t}\dot{\psi} = {} & \frac{1}{V'\langle R^{-2}\rangle}\frac{\partial}{\partial\rho}\left[V'\left\langle\frac{|\nabla\rho|^2}{R^2}\right\rangle\frac{\partial}{\partial\rho}\psi\right] \\ & - \frac{\mu_0}{\langle R^{-2}\rangle}\left(-en_\mathrm{e}\left\langle\frac{u_{\mathrm{e}\zeta}}{R}\right\rangle + Z_\mathrm{i}en_\mathrm{i}\left\langle\frac{u_{\mathrm{i}\zeta}}{R}\right\rangle + Z_\mathrm{z}en_\mathrm{z}\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle + Z_\mathrm{b}en_\mathrm{b}\left\langle\frac{u_{\mathrm{b}\zeta}}{R}\right\rangle\right)\end{aligned} \tag{1.3}$$

$$\frac{\partial}{\partial t}\psi = \dot{\psi} \tag{1.4}$$

$$\frac{\partial}{\partial t}\psi_t = \dot{\psi}_t \tag{1.5}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V'n_\mathrm{e}) = {} & -\frac{1}{V'}\frac{\partial}{\partial\rho}[V'n_\mathrm{e}(u_\mathrm{e}^\rho - u_g^\rho)] + \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03}) \\ & + Z_\mathrm{z}n_\mathrm{e}\bar{n}_z f_5 S_5 - Z_\mathrm{z}n_\mathrm{e}n_\mathrm{z}A_6 - \nu_\mathrm{L}(n_\mathrm{e} - n_\mathrm{esup}) + (1 - f_\mathrm{CX})\frac{P_\mathrm{b}}{E_\mathrm{b}}\end{aligned} \tag{1.6}$$

$$\begin{aligned}0 = {} & -(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial\rho}n_\mathrm{e}T_\mathrm{e} + en_\mathrm{e}(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial\rho}\Phi \\ & - e\frac{\partial\psi}{\partial\rho}In_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle + e\frac{\partial\psi}{\partial\rho}\langle B^2\rangle n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle\end{aligned} \tag{1.7}$$

$$\begin{aligned}m_\mathrm{e}n_\mathrm{e}\frac{\partial}{\partial t}\langle Bu_{\mathrm{e}\parallel}\rangle = {} & -\hat{\mu}_1^\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle - \hat{\mu}_2^\mathrm{e}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle + \hat{\mu}_1^\mathrm{e}BV_{1\mathrm{e}} + \hat{\mu}_2^\mathrm{e}BV_{2\mathrm{e}} \\ & + \ell_{11}^{\mathrm{ee}}\langle Bu_{\mathrm{e}\parallel}\rangle + \ell_{11}^{\mathrm{ei}}\langle Bu_{\mathrm{i}\parallel}\rangle + \ell_{11}^{\mathrm{ez}}\langle Bu_{\mathrm{z}\parallel}\rangle + \ell_{11}^{\mathrm{eb}}\langle Bu_{\mathrm{b}\parallel}\rangle \\ & - \ell_{12}^{\mathrm{ee}}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle - \ell_{12}^{\mathrm{ei}}\langle B\hat{q}_{\mathrm{i}\parallel}\rangle - \ell_{12}^{\mathrm{ez}}\langle B\hat{q}_{\mathrm{z}\parallel}\rangle \\ & - en_\mathrm{e}\langle BE_\parallel\rangle - \nu_\mathrm{L}m_\mathrm{e}n_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle - \nu_{0\mathrm{e}}m_\mathrm{e}n_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle \\ & - \frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_\mathrm{e}n_\mathrm{e}\langle Bu_{\mathrm{e}\parallel}\rangle + \langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_\mathrm{e}n_\mathrm{e}\left(\frac{\partial}{\partial\rho}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\parallel}\rangle\end{aligned}$$

$$
\begin{aligned}
&+\frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial\rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\parallel}\rangle\\
&-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\left(\frac{\partial}{\partial\rho}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial\rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\parallel}\rangle-\frac{\langle B^2\rangle}{I}\frac{1}{V'}\frac{\partial}{\partial\rho}V'\langle\Pi_{\mathrm{e}}^{\mathrm{res}}\rangle\\
&-\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\left(\langle R^2\rangle-\frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I}BV_{1\mathrm{e}}\right.\\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\partial}{\partial\rho}\left(\langle R^2\rangle-\frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I}BV_{1\mathrm{e}}\right] \qquad (1.8)
\end{aligned}
$$

$$
\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V'm_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle)=&-\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle+(u_{\mathrm{e}}^{\rho}-u_g^{\rho})m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle\right.\\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{e}\zeta}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\partial}{\partial\rho}\langle Ru_{\mathrm{e}\zeta}\rangle+\langle\Pi_{\mathrm{e}}^{\mathrm{res}}\rangle\right]\\
&+\ell_{11}^{\mathrm{ee}}\langle Ru_{\mathrm{e}\zeta}\rangle+\ell_{11}^{\mathrm{ei}}\langle Ru_{\mathrm{i}\zeta}\rangle+\ell_{11}^{\mathrm{ez}}\langle Ru_{\mathrm{z}\zeta}\rangle+\ell_{11}^{\mathrm{eb}}\langle Ru_{\mathrm{b}\zeta}\rangle\\
&-\ell_{12}^{\mathrm{ee}}\langle R\hat{q}_{\mathrm{e}\zeta}\rangle-\ell_{12}^{\mathrm{ei}}\langle R\hat{q}_{\mathrm{i}\zeta}\rangle-\ell_{12}^{\mathrm{ez}}\langle R\hat{q}_{\mathrm{z}\zeta}\rangle\\
&-en_{\mathrm{e}}\langle RE_{\zeta}\rangle-e\frac{\partial\psi}{\partial\rho}n_{\mathrm{e}}u_{\mathrm{e}}^{\rho}\\
&-\frac{e\langle|\nabla\rho|^2\rangle D_{\mathrm{e}}}{T_{\mathrm{e}}}\frac{\partial\psi}{\partial\rho}\left[(1-C_{\mathrm{T}})T_{\mathrm{e}}\frac{\partial}{\partial\rho}n_{\mathrm{e}}+C_{\mathrm{T}}\frac{\partial}{\partial\rho}n_{\mathrm{e}}T_{\mathrm{e}}+n_{\mathrm{e}}T_{\mathrm{e}}\frac{\partial\psi}{\partial\rho}C_{\mathrm{p}}\right]\\
&-\nu_{\mathrm{L}}m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle-\nu_{0\mathrm{e}}m_{\mathrm{e}}n_{\mathrm{e}}\langle Ru_{\mathrm{e}\zeta}\rangle \qquad (1.9)
\end{aligned}
$$

$$
\begin{aligned}
\frac{3}{2}\frac{1}{V'^{5/3}}\frac{\partial}{\partial t}(V'^{5/3}n_{\mathrm{e}}T_{\mathrm{e}})=&-\frac{1}{V'}\frac{\partial}{\partial\rho}V'\left[-\chi_{\mathrm{e}}\langle|\nabla\rho|^2\rangle n_{\mathrm{e}}\frac{\partial}{\partial\rho}T_{\mathrm{e}}+V_{\mathrm{e}}^{\mathrm{hp}}\langle|\nabla\rho|\rangle n_{\mathrm{e}}T_{\mathrm{e}}+\frac{5}{2}n_{\mathrm{e}}T_{\mathrm{e}}(u_{\mathrm{e}}^{\rho}-u_g^{\rho})\right]\\
&-u_g^{\rho}\frac{\partial}{\partial\rho}n_{\mathrm{e}}T_{\mathrm{e}}-u_{\mathrm{i}}^{\rho}\frac{\partial}{\partial\rho}n_{\mathrm{i}}T_{\mathrm{i}}-u_{\mathrm{z}}^{\rho}\frac{\partial}{\partial\rho}n_{\mathrm{z}}T_{\mathrm{z}}-\hat{u}_{\mathrm{i}\theta}\langle\boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\mathrm{i}}\rangle-\hat{u}_{\mathrm{z}\theta}\langle\boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\mathrm{z}}\rangle\\
&-\frac{3}{2}n_{\mathrm{e}}\frac{T_{\mathrm{e}}-T_{\mathrm{i}}}{\tau_{\mathrm{ei}}}-\frac{3}{2}n_{\mathrm{e}}\frac{T_{\mathrm{e}}-T_{\mathrm{z}}}{\tau_{\mathrm{ez}}}\\
&-\dot{\psi}_t\langle B^{\theta}\rangle(-en_{\mathrm{e}}\hat{u}_{\mathrm{e}\theta}+Z_{\mathrm{i}}en_{\mathrm{i}}\hat{u}_{\mathrm{i}\theta}+Z_{\mathrm{z}}en_{\mathrm{z}}\hat{u}_{\mathrm{z}\theta})\\
&+\dot{\psi}\left(-en_{\mathrm{e}}\left\langle\frac{u_{\mathrm{e}\zeta}}{R}\right\rangle+Z_{\mathrm{i}}en_{\mathrm{i}}\left\langle\frac{u_{\mathrm{i}\zeta}}{R}\right\rangle+Z_{\mathrm{z}}en_{\mathrm{z}}\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle\right)\\
&-\nu_{\mathrm{L}}T_{\mathrm{e}}(n_{\mathrm{e}}-n_{\mathrm{esup}})-\frac{3}{2}\nu_{\mathrm{L}T_{\mathrm{e}}}n_{\mathrm{e}}(T_{\mathrm{e}}-T_{\mathrm{esup}})\\
&-E_{\mathrm{H}}\nu_{\mathrm{ionize}}\frac{n_{\mathrm{e}}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03})\\
&+f_{\mathrm{ei}}P_{\mathrm{b}}+P_{\alpha\mathrm{e}}+P_{\mathrm{RFe}}-P_{\mathrm{Br}} \qquad (1.10)
\end{aligned}
$$

$$
\frac{5}{2}m_{\mathrm{e}}n_{\mathrm{e}}\frac{\partial}{\partial t}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle=-\hat{\mu}_2^{\mathrm{e}}\langle Bu_{\mathrm{e}\parallel}\rangle-\hat{\mu}_3^{\mathrm{e}}\langle B\hat{q}_{\mathrm{e}\parallel}\rangle+\hat{\mu}_2^{\mathrm{e}}BV_{1\mathrm{e}}+\hat{\mu}_3^{\mathrm{e}}BV_{2\mathrm{e}}
$$

$$
\begin{aligned}
&- \ell_{21}^{\mathrm{ee}}\langle Bu_{\mathrm{e}\|}\rangle - \ell_{21}^{\mathrm{ei}}\langle Bu_{\mathrm{i}\|}\rangle - \ell_{21}^{\mathrm{ez}}\langle Bu_{\mathrm{z}\|}\rangle - \ell_{21}^{\mathrm{eb}}\langle Bu_{\mathrm{b}\|}\rangle \\
&+ \ell_{22}^{\mathrm{ee}}\langle B\hat{q}_{\mathrm{e}\|}\rangle + \ell_{22}^{\mathrm{ei}}\langle B\hat{q}_{\mathrm{i}\|}\rangle + \ell_{22}^{\mathrm{ez}}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&- \nu_{\mathrm{L}} m_{\mathrm{e}} n_{\mathrm{e}}\langle B\hat{q}_{\mathrm{e}\|}\rangle - \nu_{0\mathrm{e}} m_{\mathrm{e}} n_{\mathrm{e}}\langle B\hat{q}_{\mathrm{e}\|}\rangle
\end{aligned} \tag{1.11}
$$

$$
\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{\mathrm{i}}) = &-\frac{1}{V'}\frac{\partial}{\partial \rho}[V' n_{\mathrm{i}}(u_{\mathrm{i}}^{\rho} - u_g^{\rho})] + \nu_{\mathrm{ionize}}\frac{n_{\mathrm{e}}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03}) \\
&- \nu_{\mathrm{L}}(n_{\mathrm{i}} - n_{\mathrm{isup}}) + \nu_{\mathrm{b}} n_{\mathrm{b}} + \nu_{\mathrm{b}} g n_{\mathrm{b}}^{\mathrm{rp}} - f_{\mathrm{CX}}\frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} + \nu_{\mathrm{bL}} n_{\mathrm{b}} + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{i}}
\end{aligned} \tag{1.12}
$$

$$
\begin{aligned}
0 = &-(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial \rho} n_{\mathrm{i}} T_{\mathrm{i}} - Z_{\mathrm{i}} e n_{\mathrm{i}}(\langle B^2\rangle\langle R^2\rangle - I^2)\frac{\partial}{\partial \rho}\Phi \\
&+ Z_{\mathrm{i}} e n_{\mathrm{i}}\frac{\partial \psi}{\partial \rho} I\langle Bu_{\mathrm{i}\|}\rangle - Z_{\mathrm{i}} e n_{\mathrm{i}}\frac{\partial \psi}{\partial \rho}\langle B^2\rangle\langle Ru_{\mathrm{i}\zeta}\rangle
\end{aligned} \tag{1.13}
$$

$$
\begin{aligned}
m_{\mathrm{i}} n_{\mathrm{i}}\frac{\partial}{\partial t}\langle Bu_{\mathrm{i}\|}\rangle = &-\hat{\mu}_1^{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle - \hat{\mu}_2^{\mathrm{i}}\langle B\hat{q}_{\mathrm{i}\|}\rangle + \hat{\mu}_1^{\mathrm{i}} BV_{1\mathrm{i}} + \hat{\mu}_2^{\mathrm{i}} BV_{2\mathrm{i}} \\
&+ \ell_{11}^{\mathrm{ie}}\langle Bu_{\mathrm{e}\|}\rangle + \ell_{11}^{\mathrm{ii}}\langle Bu_{\mathrm{i}\|}\rangle + \ell_{11}^{\mathrm{iz}}\langle Bu_{\mathrm{z}\|}\rangle + \ell_{11}^{\mathrm{ib}}\langle Bu_{\mathrm{b}\|}\rangle \\
&- \ell_{12}^{\mathrm{ie}}\langle B\hat{q}_{\mathrm{e}\|}\rangle - \ell_{12}^{\mathrm{ii}}\langle B\hat{q}_{\mathrm{i}\|}\rangle - \ell_{12}^{\mathrm{iz}}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&+ Z_{\mathrm{i}} e n_{\mathrm{i}}\langle BE_{\|}\rangle - \nu_{\mathrm{L}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle + \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}}\langle Bu_{\mathrm{b}\|}\rangle - \nu_{0\mathrm{i}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle \\
&- \nu_{\mathrm{CX}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle - \nu_{\mathrm{OL}} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle \\
&- \frac{1}{V'}\frac{\partial}{\partial \rho} V'\langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\langle Bu_{\mathrm{i}\|}\rangle + \langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\left(\frac{\partial}{\partial \rho}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{i}\|}\rangle \\
&+ \frac{1}{V'}\frac{\partial}{\partial \rho} V'\langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial \rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{i}\|}\rangle \\
&- \langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\left(\frac{\partial}{\partial \rho}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial \rho}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{i}\|}\rangle - \frac{\langle B^2\rangle}{I}\frac{1}{V'}\frac{\partial}{\partial \rho} V'\langle\Pi_{\mathrm{i}}^{\mathrm{res}}\rangle \\
&- \frac{1}{V'}\frac{\partial}{\partial \rho} V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\left(\langle R^2\rangle - \frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I} BV_{1\mathrm{i}}\right. \\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\frac{\partial}{\partial \rho}\left(\langle R^2\rangle - \frac{I^2}{\langle B^2\rangle}\right)\frac{1}{I} BV_{1\mathrm{i}}\right]
\end{aligned} \tag{1.14}
$$

$$
\begin{aligned}
\frac{1}{V'}\frac{\partial}{\partial t}(V' m_{\mathrm{i}} n_{\mathrm{i}}\langle Ru_{\mathrm{i}\zeta}\rangle) = &-\frac{1}{V'}\frac{\partial}{\partial \rho} V'\left[\langle|\nabla\rho|\rangle v_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\langle Ru_{\mathrm{i}\zeta}\rangle + (u_{\mathrm{i}}^{\rho} - u_g^{\rho}) m_{\mathrm{i}} n_{\mathrm{i}}\langle Ru_{\mathrm{i}\zeta}\rangle\right. \\
&\left.-\langle|\nabla\rho|^2\rangle\chi_{\mathrm{i}\zeta} m_{\mathrm{i}} n_{\mathrm{i}}\frac{\partial}{\partial \rho}\langle Ru_{\mathrm{i}\zeta}\rangle + \langle\Pi_{\mathrm{i}}^{\mathrm{res}}\rangle\right] \\
&+ \ell_{11}^{\mathrm{ie}}\langle Ru_{\mathrm{e}\zeta}\rangle + \ell_{11}^{\mathrm{ii}}\langle Ru_{\mathrm{i}\zeta}\rangle + \ell_{11}^{\mathrm{iz}}\langle Ru_{\mathrm{z}\zeta}\rangle + \ell_{11}^{\mathrm{ib}}\langle Ru_{\mathrm{b}\zeta}\rangle
\end{aligned}
$$

$$
\begin{aligned}
&- \ell_{12}^{\mathrm{ie}}\langle R\hat{q}_{\mathrm{e}\zeta}\rangle - \ell_{12}^{\mathrm{ii}}\langle R\hat{q}_{\mathrm{i}\zeta}\rangle - \ell_{12}^{\mathrm{iz}}\langle R\hat{q}_{\mathrm{z}\zeta}\rangle \\
&+ Z_{\mathrm{i}} e n_{\mathrm{i}}\langle RE_{\zeta}\rangle + Z_{\mathrm{i}} e \frac{\partial \psi}{\partial \rho} n_{\mathrm{i}} u_{\mathrm{i}}^{\rho} \\
&+ Z_{\mathrm{i}} \frac{e\langle |\nabla\rho|^2\rangle D_{\mathrm{e}}}{T_{\mathrm{e}}} \frac{\partial \psi}{\partial \rho} \frac{n_{\mathrm{i}} + n_{\mathrm{b}}}{n_{\mathrm{e}}} \left[(1 - C_{\mathrm{T}}) T_{\mathrm{e}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} + C_{\mathrm{T}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} T_{\mathrm{e}} + n_{\mathrm{e}} T_{\mathrm{e}} \frac{\partial \psi}{\partial \rho} C_{\mathrm{p}} \right] \\
&- \nu_{\mathrm{L}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle + \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta}\rangle - \nu_{0\mathrm{i}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle - \nu_{\mathrm{CX}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle \\
&- \nu_{\mathrm{OL}} m_{\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta}\rangle
\end{aligned} \tag{1.15}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{1}{V'^{5/3}} \frac{\partial}{\partial t} (V'^{5/3} n_{\mathrm{i}} T_{\mathrm{i}}) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[-\chi_{\mathrm{i}} \langle |\nabla\rho|^2\rangle n_{\mathrm{i}} \frac{\partial}{\partial \rho} T_{\mathrm{i}} + V_{\mathrm{i}}^{\mathrm{hp}} \langle |\nabla\rho|\rangle n_{\mathrm{i}} T_{\mathrm{i}} + \frac{5}{2} n_{\mathrm{i}} T_{\mathrm{i}} (u_{\mathrm{i}}^{\rho} - u_g^{\rho}) \right] \\
&+ (u_{\mathrm{i}}^{\rho} - u_g^{\rho}) \frac{\partial}{\partial \rho} n_{\mathrm{i}} T_{\mathrm{i}} + \hat{u}_{\mathrm{i}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{i}} \rangle - \frac{3}{2} n_{\mathrm{i}} \frac{T_{\mathrm{i}} - T_{\mathrm{e}}}{\tau_{\mathrm{ie}}} - \frac{3}{2} n_{\mathrm{i}} \frac{T_{\mathrm{i}} - T_{\mathrm{z}}}{\tau_{\mathrm{iz}}} \\
&- \nu_{\mathrm{L}} T_{\mathrm{i}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) - \frac{3}{2} \nu_{\mathrm{L}T_{\mathrm{i}}} n_{\mathrm{i}} (T_{\mathrm{i}} - T_{\mathrm{isup}}) \\
&+ \frac{3}{2} \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} T_{01} + n_{02} T_{02} + n_{03} T_{03}) \\
&- \frac{3}{2} \nu_{\mathrm{CX}} n_{\mathrm{i}} (T_{\mathrm{i}} - T_{01}) + (1 - f_{\mathrm{ei}}) P_{\mathrm{b}} + m_{\mathrm{b}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} \langle B u_{\mathrm{b}\parallel}\rangle \frac{\langle B u_{\mathrm{i}\parallel}\rangle}{\langle B^2\rangle} + P_{\alpha\mathrm{i}} + P_{\mathrm{RFi}}
\end{aligned} \tag{1.16}
$$

$$
\begin{aligned}
\frac{5}{2} m_{\mathrm{i}} n_{\mathrm{i}} \frac{\partial}{\partial t} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle = &- \hat{\mu}_2^{\mathrm{i}} \langle B u_{\mathrm{i}\parallel}\rangle - \hat{\mu}_3^{\mathrm{i}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle + \hat{\mu}_2^{\mathrm{i}} B V_{1\mathrm{i}} + \hat{\mu}_3^{\mathrm{i}} B V_{2\mathrm{i}} \\
&- \ell_{21}^{\mathrm{ie}} \langle B u_{\mathrm{e}\parallel}\rangle - \ell_{21}^{\mathrm{ii}} \langle B u_{\mathrm{i}\parallel}\rangle - \ell_{21}^{\mathrm{iz}} \langle B u_{\mathrm{z}\parallel}\rangle - \ell_{21}^{\mathrm{ib}} \langle B u_{\mathrm{b}\parallel}\rangle \\
&+ \ell_{22}^{\mathrm{ie}} \langle B\hat{q}_{\mathrm{e}\parallel}\rangle + \ell_{22}^{\mathrm{ii}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle + \ell_{22}^{\mathrm{iz}} \langle B\hat{q}_{\mathrm{z}\parallel}\rangle \\
&- \nu_{\mathrm{L}} m_{\mathrm{i}} n_{\mathrm{i}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle - \nu_{0\mathrm{i}} m_{\mathrm{i}} n_{\mathrm{i}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle
\end{aligned} \tag{1.17}
$$

$$
\begin{aligned}
\frac{1}{V'} \frac{\partial}{\partial t} (V' n_{\mathrm{z}}) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} [V' n_{\mathrm{z}} (u_{\mathrm{z}}^{\rho} - u_g^{\rho})] + n_{\mathrm{e}} \bar{n}_z f_5 S_5 - n_{\mathrm{e}} n_{\mathrm{z}} A_6 \\
&- \nu_{\mathrm{L}} (n_{\mathrm{z}} - n_{\mathrm{zsup}}) + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{z}}
\end{aligned} \tag{1.18}
$$

$$
\begin{aligned}
0 = &- (\langle B^2\rangle \langle R^2\rangle - I^2) \frac{\partial}{\partial \rho} n_{\mathrm{z}} T_{\mathrm{z}} - Z_{\mathrm{z}} e n_{\mathrm{z}} (\langle B^2\rangle \langle R^2\rangle - I^2) \frac{\partial}{\partial \rho} \Phi \\
&+ Z_{\mathrm{z}} e n_{\mathrm{z}} \frac{\partial \psi}{\partial \rho} I \langle B u_{\mathrm{z}\parallel}\rangle - Z_{\mathrm{z}} e n_{\mathrm{z}} \frac{\partial \psi}{\partial \rho} \langle B^2\rangle \langle R u_{\mathrm{z}\zeta}\rangle
\end{aligned} \tag{1.19}
$$

$$
\begin{aligned}
m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial t} \langle B u_{\mathrm{z}\parallel}\rangle = &- \hat{\mu}_1^{\mathrm{z}} \langle B u_{\mathrm{z}\parallel}\rangle - \hat{\mu}_2^{\mathrm{z}} \langle B\hat{q}_{\mathrm{z}\parallel}\rangle + \hat{\mu}_1^{\mathrm{z}} B V_{1\mathrm{z}} + \hat{\mu}_2^{\mathrm{z}} B V_{2\mathrm{z}} \\
&+ \ell_{11}^{\mathrm{ze}} \langle B u_{\mathrm{e}\parallel}\rangle + \ell_{11}^{\mathrm{zi}} \langle B u_{\mathrm{i}\parallel}\rangle + \ell_{11}^{\mathrm{zz}} \langle B u_{\mathrm{z}\parallel}\rangle + \ell_{11}^{\mathrm{zb}} \langle B u_{\mathrm{b}\parallel}\rangle \\
&- \ell_{12}^{\mathrm{ze}} \langle B\hat{q}_{\mathrm{e}\parallel}\rangle - \ell_{12}^{\mathrm{zi}} \langle B\hat{q}_{\mathrm{i}\parallel}\rangle - \ell_{12}^{\mathrm{zz}} \langle B\hat{q}_{\mathrm{z}\parallel}\rangle
\end{aligned}
$$

$$
\begin{aligned}
&+ Z_{\mathrm{z}} e n_{\mathrm{z}} \langle B E_{\parallel} \rangle - \nu_{\mathrm{L}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{0\mathrm{z}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{\mathrm{OL}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle \\
&- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle + \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \left(\frac{\partial}{\partial \rho} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
&+ \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial \rho} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
&- \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \left(\frac{\partial}{\partial \rho} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial \rho} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle - \frac{\langle B^2 \rangle}{I} \frac{1}{V'} \frac{\partial}{\partial \rho} V' \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \\
&- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[\langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \left(\langle R^2 \rangle - \frac{I^2}{\langle B^2 \rangle} \right) \frac{1}{I} B V_{1\mathrm{z}} \right. \\
&\left. - \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial \rho} \left(\langle R^2 \rangle - \frac{I^2}{\langle B^2 \rangle} \right) \frac{1}{I} B V_{1\mathrm{z}} \right] \qquad (1.20)
\end{aligned}
$$

$$
\begin{aligned}
\frac{1}{V'} \frac{\partial}{\partial t} (V' m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[\langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle + (u_{\mathrm{z}}^{\rho} - u_g^{\rho}) m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \right. \\
&\left. - \langle |\nabla \rho|^2 \rangle \chi_{\mathrm{z}\zeta} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial \rho} \langle R u_{\mathrm{z}\zeta} \rangle + \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \right] \\
&+ \ell_{11}^{\mathrm{ze}} \langle R u_{\mathrm{e}\zeta} \rangle + \ell_{11}^{\mathrm{zi}} \langle R u_{\mathrm{i}\zeta} \rangle + \ell_{11}^{\mathrm{zz}} \langle R u_{\mathrm{z}\zeta} \rangle + \ell_{11}^{\mathrm{zb}} \langle R u_{\mathrm{b}\zeta} \rangle \\
&- \ell_{12}^{\mathrm{ze}} \langle R \hat{q}_{\mathrm{e}\zeta} \rangle - \ell_{12}^{\mathrm{zi}} \langle R \hat{q}_{\mathrm{i}\zeta} \rangle - \ell_{12}^{\mathrm{zz}} \langle R \hat{q}_{\mathrm{z}\zeta} \rangle \\
&+ Z_{\mathrm{z}} e n_{\mathrm{z}} \langle R E_{\zeta} \rangle + Z_{\mathrm{z}} e \frac{\partial \psi}{\partial \rho} n_{\mathrm{z}} u_{\mathrm{z}}^{\rho} \\
&+ Z_{\mathrm{z}} \frac{e \langle |\nabla \rho|^2 \rangle D_{\mathrm{e}}}{T_{\mathrm{e}}} \frac{\partial \psi}{\partial \rho} \frac{n_{\mathrm{z}}}{n_{\mathrm{e}}} \left[(1 - C_{\mathrm{T}}) T_{\mathrm{e}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} + C_{\mathrm{T}} \frac{\partial}{\partial \rho} n_{\mathrm{e}} T_{\mathrm{e}} + n_{\mathrm{e}} T_{\mathrm{e}} \frac{\partial \psi}{\partial \rho} C_{\mathrm{p}} \right] \\
&- \nu_{\mathrm{L}} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \nu_{0\mathrm{z}} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \nu_{\mathrm{OL}} m_{\mathrm{z}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \qquad (1.21)
\end{aligned}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{1}{V'^{5/3}} \frac{\partial}{\partial t} (V'^{5/3} n_{\mathrm{z}} T_{\mathrm{z}}) = &- \frac{1}{V'} \frac{\partial}{\partial \rho} V' \left[- \chi_{\mathrm{z}} \langle |\nabla \rho|^2 \rangle n_{\mathrm{z}} \frac{\partial}{\partial \rho} T_{\mathrm{z}} + V_{\mathrm{z}}^{\mathrm{hp}} \langle |\nabla \rho| \rangle n_{\mathrm{z}} T_{\mathrm{z}} + \frac{5}{2} n_{\mathrm{z}} T_{\mathrm{z}} (u_{\mathrm{z}}^{\rho} - u_g^{\rho}) \right] \\
&+ (u_{\mathrm{z}}^{\rho} - u_g^{\rho}) \frac{\partial}{\partial \rho} n_{\mathrm{z}} T_{\mathrm{z}} + \hat{u}_{\mathrm{z}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{z}} \rangle - \frac{3}{2} n_{\mathrm{z}} \frac{T_{\mathrm{z}} - T_{\mathrm{e}}}{\tau_{\mathrm{ze}}} - \frac{3}{2} n_{\mathrm{z}} \frac{T_{\mathrm{z}} - T_{\mathrm{i}}}{\tau_{\mathrm{zi}}} \\
&- \nu_{\mathrm{L}} T_{\mathrm{z}} (n_{\mathrm{z}} - n_{\mathrm{zsup}}) - \frac{3}{2} \nu_{\mathrm{L}T_{\mathrm{z}}} n_{\mathrm{z}} (T_{\mathrm{z}} - T_{\mathrm{zsup}}) \\
&+ P_{\alpha \mathrm{z}} + P_{\mathrm{RFz}} \qquad (1.22)
\end{aligned}
$$

$$
\begin{aligned}
\frac{5}{2} m_{\mathrm{z}} n_{\mathrm{z}} \frac{\partial}{\partial t} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle = &- \hat{\mu}_2^{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \hat{\mu}_3^{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle + \hat{\mu}_2^{\mathrm{z}} B V_{1\mathrm{z}} + \hat{\mu}_3^{\mathrm{z}} B V_{2\mathrm{z}} \\
&- \ell_{21}^{\mathrm{ze}} \langle B u_{\mathrm{e}\parallel} \rangle - \ell_{21}^{\mathrm{zi}} \langle B u_{\mathrm{i}\parallel} \rangle - \ell_{21}^{\mathrm{zz}} \langle B u_{\mathrm{z}\parallel} \rangle - \ell_{21}^{\mathrm{zb}} \langle B u_{\mathrm{b}\parallel} \rangle \\
&+ \ell_{22}^{\mathrm{ze}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \ell_{22}^{\mathrm{zi}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \ell_{22}^{\mathrm{zz}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle
\end{aligned}
$$

$$- \nu_{\mathrm{L}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\|} \rangle - \nu_{0\mathrm{z}} m_{\mathrm{z}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\|} \rangle \tag{1.23}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{\mathrm{b}}) = &-\frac{1}{V'}\frac{\partial}{\partial \rho}[V' n_{\mathrm{b}}(u^{\rho}_{\mathrm{b}} - u^{\rho}_{g})] + \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} - g\frac{P_{\mathrm{b}\perp}}{E_{\mathrm{b}}} - \nu_{\mathrm{b}} n_{\mathrm{b}} - \nu_{\mathrm{bL}} n_{\mathrm{b}} - \nu_{\mathrm{eff}} g(\sqrt{\delta} n_{\mathrm{b}} - n^{\mathrm{rp}}_{\mathrm{b}}) \\ &+ \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle f_t D_{\mathrm{b}} \frac{\partial}{\partial \rho} n_{\mathrm{b}}\end{aligned} \tag{1.24}$$

$$\begin{aligned}0 = &-(\langle B^2 \rangle \langle R^2 \rangle - I^2)\frac{\partial}{\partial \rho} n_{\mathrm{b}} T_{\mathrm{b}} - Z_{\mathrm{b}} e n_{\mathrm{b}} (\langle B^2 \rangle \langle R^2 \rangle - I^2)\frac{\partial}{\partial \rho}\Phi \\ &+ Z_{\mathrm{b}} e n_{\mathrm{b}} \frac{\partial \psi}{\partial \rho} I \langle B u_{\mathrm{b}\|} \rangle - Z_{\mathrm{b}} e n_{\mathrm{b}} \frac{\partial \psi}{\partial \rho} \langle B^2 \rangle \langle R u_{\mathrm{b}\zeta} \rangle\end{aligned} \tag{1.25}$$

$$\begin{aligned}m_{\mathrm{b}} \frac{\partial}{\partial t} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle = &-(\hat{\mu}^{\mathrm{b}}_1 - \ell^{\mathrm{bb}}_{11}) \langle B u_{\mathrm{b}\|} \rangle + \sum_{a \neq \mathrm{b}} \ell^{\mathrm{b}a}_{11} \langle B u_{a\|} \rangle + \hat{\mu}^{\mathrm{b}}_1 B V_{1\mathrm{b}} + Z_{\mathrm{b}} e n_{\mathrm{b}} \langle B E^{A}_{\|} \rangle \\ &- \nu_{\mathrm{bL}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle - \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle - \nu_{0\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle - \nu_{\mathrm{CX}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle + \langle B S_{\mathrm{m}\|\mathrm{b}} \rangle \\ &- \nu_{\mathrm{eff}} g \sqrt{\delta} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle + \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle m_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle f_t D_{\mathrm{b}} \frac{\partial}{\partial \rho} n_{\mathrm{b}}\end{aligned} \tag{1.26}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V' m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle) = &-\frac{1}{V'}\frac{\partial}{\partial \rho}[V'(u^{\rho}_{\mathrm{b}} - u^{\rho}_{g}) m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle] \\ &+ \ell^{\mathrm{be}}_{11} \langle R u_{\mathrm{e}\zeta} \rangle + \ell^{\mathrm{bi}}_{11} \langle R u_{\mathrm{i}\zeta} \rangle + \ell^{\mathrm{bz}}_{11} \langle R u_{\mathrm{z}\zeta} \rangle + \ell^{\mathrm{bb}}_{11} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle \\ &+ Z_{\mathrm{b}} e n_{\mathrm{b}} \dot{\psi} + Z_{\mathrm{b}} e \frac{\partial \psi}{\partial \rho} n_{\mathrm{b}} u^{\rho}_{\mathrm{b}} - \nu_{\mathrm{bL}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle - \nu_{\mathrm{b}} m_{\mathrm{b}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\zeta} \rangle \\ &- \nu_{0\mathrm{i}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle - \nu_{\mathrm{CX}} m_{\mathrm{b}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle + \frac{I}{\langle B^2 \rangle} \langle B S_{\mathrm{m}\|\mathrm{b}} \rangle\end{aligned} \tag{1.27}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{01}) = &\frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{01} \frac{\partial}{\partial \rho} n_{01} - \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} n_{01} \\ &- \nu_{\mathrm{CX}} \frac{n_{\mathrm{i}}}{n_{01} + n_{02}} n_{01} + \gamma_0 \nu_{\mathrm{L}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) + \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle \mathbf{\Gamma}^{\mathrm{puff}}_{\mathrm{i}} \cdot \nabla \rho \rangle \Big|_{\mathrm{b}}\end{aligned} \tag{1.28}$$

$$\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{02}) = \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{02} \frac{\partial}{\partial \rho} n_{02} - \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} n_{02} + \nu_{\mathrm{CX}} \frac{n_{\mathrm{i}}}{n_{01} + n_{02}} n_{01} \tag{1.29}$$

$$\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{03}) = \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{03} \frac{\partial}{\partial \rho} n_{03} - \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} n_{03} + f_{\mathrm{CX}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} \tag{1.30}$$

$$\begin{aligned}\frac{1}{V'}\frac{\partial}{\partial t}(V' \bar{n}_z) = &\frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle |\nabla \rho|^2 \rangle D_{0\mathrm{z}} \frac{\partial}{\partial \rho} \bar{n}_z - n_{\mathrm{e}} \bar{n}_z f_5 S_5 + n_{\mathrm{e}} n_{\mathrm{z}} A_6 \\ &+ \gamma_{0\mathrm{z}} \nu_{\mathrm{L}} (n_{\mathrm{z}} - n_{\mathrm{zsup}}) + \gamma_{\mathrm{i,z}} \nu_{\mathrm{L}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) + \frac{1}{V'}\frac{\partial}{\partial \rho} V' \langle \mathbf{\Gamma}^{\mathrm{puff}}_{\mathrm{z}} \cdot \nabla \rho \rangle \Big|_{\mathrm{b}}\end{aligned} \tag{1.31}$$

$$\frac{1}{V'}\frac{\partial}{\partial t}(V' n_{\rm b}^{\rm rp}) = \frac{P_{\rm b\perp}}{E_{\rm b}} + \nu_{\rm eff}(\sqrt{\delta} n_{\rm b} - n_{\rm b}^{\rm rp}) - \nu_{\rm b} n_{\rm b}^{\rm rp} - \frac{1}{V'}\frac{\partial}{\partial \rho} V' \left(\langle |\nabla \rho| \rangle u_{\rm b}^{\rm rp} n_{\rm b}^{\rm rp} - \langle |\nabla \rho|^2 \rangle D_{\rm b}^{\rm rp} \frac{\partial}{\partial \rho} n_{\rm b}^{\rm rp} \right) \quad (1.32)$$

2 Additional important relationship

$$\bar{n}_z = \sum_{i=0}^{\max(Z)-1} n_i \quad (2.1)$$

$$B_p = \frac{|\nabla \rho|}{R}\frac{\partial \psi}{\partial \rho}, \quad \langle B_p^2 \rangle = \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left(\frac{\partial \psi}{\partial V} \right)^2 \quad (2.2)$$

$$B_t = \frac{I}{R} = \frac{1}{R}\frac{4\pi^2}{\langle R^{-2} \rangle}\frac{\partial \psi_t}{\partial V}, \quad \langle B_t^2 \rangle = \frac{16\pi^4}{\langle R^{-2} \rangle}\left(\frac{\partial \psi_t}{\partial V} \right)^2 \quad (2.3)$$

$$\langle B^2 \rangle = \langle B_t^2 \rangle + \langle B_p^2 \rangle = \frac{16\pi^4}{\langle R^{-2} \rangle}\left(\frac{\partial \psi_t}{\partial V} \right)^2 + \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left(\frac{\partial \psi}{\partial V} \right)^2 \quad (2.4)$$

$$\langle B^\theta \rangle = \frac{4\pi^2}{V'}\frac{\partial \psi}{\partial \rho} = 4\pi^2 \frac{\partial \psi}{\partial V} = \frac{I \langle R^{-2} \rangle}{q} = 2\pi \left(\oint \frac{dl_p}{B_p} \right)^{-1} \quad (2.5)$$

$$\hat{q}^2 \equiv \frac{I^2}{2\langle B_p^2 \rangle}\left(\left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right) \quad (2.6)$$

$$\langle B^2 \rangle \langle R^2 \rangle - I^2 = \langle R^2 \rangle \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \left(\frac{\partial \psi}{\partial V} \right)^2 (1 + 2\hat{q}^2) \quad (2.7)$$

$$\langle B \hat{q}_{s\|} \rangle \equiv \frac{2\langle B q_{s\|} \rangle}{5 p_s} \quad (2.8)$$

$$\hat{u}_{s\theta} = \frac{\langle B u_{s\|} \rangle - B V_{1s}}{\langle B^2 \rangle} \quad (2.9)$$

$$\begin{aligned} \left\langle \frac{u_{s\zeta}}{R} \right\rangle &= \frac{1}{1+2\hat{q}^2}\frac{1}{\langle R^2 \rangle}\langle R u_{s\zeta} \rangle + \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{I}\langle B u_{s\|} \rangle \quad (2.10) \\ &= \frac{\langle R u_{s\zeta} \rangle}{\langle R^2 \rangle} + \frac{I}{\langle B^2 \rangle}\left(\left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right)(\langle B u_{s\|} \rangle - B V_{1s}) \\ &= \frac{\langle R u_{s\zeta} \rangle}{\langle R^2 \rangle} + I \left(\left\langle \frac{1}{R^2} \right\rangle - \frac{1}{\langle R^2 \rangle} \right) \hat{u}_{s\theta} \end{aligned}$$

$$\langle R \hat{q}_{s\zeta} \rangle = \frac{1}{\langle B^2 \rangle}\left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right)\frac{1}{I} B V_{2s} + \frac{I}{\langle B^2 \rangle}\langle B \hat{q}_{s\|} \rangle \quad (2.11)$$

$$B V_{1s} = -I \left(\frac{1}{e_s n_s}\frac{\partial n_s T_s}{\partial \psi} + \frac{\partial \Phi}{\partial \psi} \right), \quad B V_{2s} = -\frac{I}{e_s}\frac{\partial T_s}{\partial \psi} \quad (2.12)$$

$$u^\rho \equiv \langle \boldsymbol{u}\cdot\nabla\rho\rangle, \quad u^V = \langle \boldsymbol{u}\cdot\nabla V\rangle \tag{2.13}$$

$$u^\rho_g = -\frac{4\pi^2}{IV'\langle R^{-2}\rangle}\dot{\psi}_t, \quad u^V_g = -\frac{4\pi^2}{I\langle R^{-2}\rangle}\dot{\psi}_t = -\frac{V'}{2\rho\psi_{ta}}\dot{\psi}_t \tag{2.14}$$

$$\frac{1}{V'}\frac{\partial V'}{\partial t} = \frac{\partial}{\partial V}\langle \boldsymbol{u}_g\cdot\nabla V\rangle = \frac{\partial u^V_g}{\partial V} \tag{2.15}$$

$$\langle E_r\rangle = -\langle|\nabla\rho|^2\rangle\frac{\partial r}{\partial\rho}\frac{\partial}{\partial\rho}\Phi \tag{2.16}$$

$$\langle BE_\parallel\rangle = I\langle R^{-2}\rangle\left(-\frac{\dot{\psi}_t}{q}+\dot{\psi}\right) \tag{2.17}$$

$$RE_\zeta = \dot{\psi} = \frac{V_{\rm loop}}{2\pi} \tag{2.18}$$

$$\langle \boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\rm i}\rangle = \hat{\mu}^{\rm i}_1(\langle Bu_{\rm i\parallel}\rangle - BV_{1\rm i}) + \hat{\mu}^{\rm i}_2\left(\frac{2\langle Bq_{\rm i\parallel}\rangle}{5p_{\rm i}} - BV_{2\rm i}\right) \tag{2.19}$$

$$\hat{\ell}^{ab}_{ij} \equiv \frac{\ell^{ab}_{ij}}{m_a n_a}, \quad \hat{\ell}^{\rm bb}_{ij} \equiv \frac{\ell^{\rm bb}_{ij}}{m_{\rm e}n_{\rm e}} \tag{2.20}$$

$$\Gamma^{\rm puff} \simeq \frac{\langle \boldsymbol{\Gamma}^{\rm puff}\cdot\nabla\rho\rangle}{\langle|\nabla\rho|\rangle} \tag{2.21}$$

3 Equations in the coordinates suitable for implementation

$$0 = \frac{\partial}{\partial V}\left[\langle|\nabla V|^2\rangle\frac{\partial}{\partial V}\Phi\right] + \frac{1}{\epsilon_0}\left(-en_{\rm e} + Z_{\rm i}en_{\rm i} + Z_{\rm z}en_{\rm z} + Z_{\rm b}en_{\rm b} + Z_{\rm b}eg n^{\rm rp}_{\rm b}\right) \tag{3.1}$$

$$\begin{aligned}\epsilon_0\frac{\partial}{\partial t}\dot{\psi}_t = {}& \frac{\partial}{\partial V}\left[\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{1}{\langle R^{-2}\rangle}\frac{\partial}{\partial V}\frac{\psi_t}{\mu_0}\right] - \frac{1}{\langle R^{-2}\rangle}\frac{\partial}{\partial V}\left[\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\right]\frac{\partial}{\partial V}\frac{\psi_t}{\mu_0}\\ & + \frac{1}{\langle B^\theta\rangle}\left[-e\left(n_{\rm e}\langle Bu_{\rm e\parallel}\rangle - In_{\rm e}\left\langle\frac{u_{\rm e\zeta}}{R}\right\rangle\right) + Z_{\rm i}e\left(n_{\rm i}\langle Bu_{\rm i\parallel}\rangle - In_{\rm i}\left\langle\frac{u_{\rm i\zeta}}{R}\right\rangle\right)\right.\\ & \left. + Z_{\rm z}e\left(n_{\rm z}\langle Bu_{\rm z\parallel}\rangle - In_{\rm z}\left\langle\frac{u_{\rm z\zeta}}{R}\right\rangle\right) + Z_{\rm b}e\left(n_{\rm b}\langle Bu_{\rm b\parallel}\rangle - In_{\rm b}\left\langle\frac{u_{\rm b\zeta}}{R}\right\rangle\right)\right]\end{aligned} \tag{3.2}$$

$$\begin{aligned}\frac{1}{c^2}\left\langle\frac{1}{R^2}\right\rangle\frac{\partial}{\partial t}\dot{\psi} = {}& \frac{\partial}{\partial V}\left[\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{\partial}{\partial V}\psi\right]\\ & - \mu_0\left(-en_{\rm e}\left\langle\frac{u_{\rm e\zeta}}{R}\right\rangle + Z_{\rm i}en_{\rm i}\left\langle\frac{u_{\rm i\zeta}}{R}\right\rangle + Z_{\rm z}en_{\rm z}\left\langle\frac{u_{\rm z\zeta}}{R}\right\rangle + Z_{\rm b}en_{\rm b}\left\langle\frac{u_{\rm b\zeta}}{R}\right\rangle\right)\end{aligned} \tag{3.3}$$

$$\frac{\partial}{\partial t}\psi = \dot{\psi} \tag{3.4}$$

$$\frac{\partial}{\partial t}\frac{\psi_t}{\mu_0} = \frac{\dot{\psi}_t}{\mu_0} \tag{3.5}$$

$$\begin{aligned}\frac{\partial}{\partial t}n_\mathrm{e} = &-\frac{\partial}{\partial V}n_\mathrm{e}u_\mathrm{e}^V + u_g^V\frac{\partial}{\partial V}n_\mathrm{e} + \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}(n_{01}+n_{02}+n_{03})\\ &+ Z_\mathrm{z}n_\mathrm{e}\bar{n}_z f_5 S_5 - Z_\mathrm{z}n_\mathrm{e}n_\mathrm{z}A_6 - \nu_\mathrm{L}(n_\mathrm{e}-n_\mathrm{esup}) + (1-f_\mathrm{CX})\frac{P_\mathrm{b}}{E_\mathrm{b}}\end{aligned} \tag{3.6}$$

$$0 = -\frac{e}{m_\mathrm{e}}n_\mathrm{e}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{1}{I}BV_{1\mathrm{e}} - \frac{e}{m_\mathrm{e}}In_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle + \frac{e}{m_\mathrm{e}}\langle B^2\rangle n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle \tag{3.7}$$

$$\begin{aligned}n_\mathrm{e}\frac{\partial}{\partial t}\langle Bu_{\mathrm{e}\|}\rangle = &-\frac{1}{m_\mathrm{e}}\hat{\mu}_1^\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle - \frac{1}{m_\mathrm{e}}\hat{\mu}_2^\mathrm{e}\langle B\hat{q}_{\mathrm{e}\|}\rangle + \frac{1}{m_\mathrm{e}}\hat{\mu}_1^\mathrm{e}BV_{1\mathrm{e}}\\ &+ \frac{1}{em_\mathrm{e}}\hat{\mu}_2^\mathrm{e}I\frac{\partial V}{\partial\psi}\frac{1}{n_\mathrm{e}}\frac{\partial}{\partial V}n_\mathrm{e}T_\mathrm{e} - \frac{1}{em_\mathrm{e}}\hat{\mu}_2^\mathrm{e}I\frac{\partial V}{\partial\psi}\frac{T_\mathrm{e}}{n_\mathrm{e}}\frac{\partial}{\partial V}n_\mathrm{e}\\ &+ \hat{\ell}_{11}^{\mathrm{ee}}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle + \hat{\ell}_{11}^{\mathrm{ei}}n_\mathrm{e}\langle Bu_{\mathrm{i}\|}\rangle + \hat{\ell}_{11}^{\mathrm{ez}}n_\mathrm{e}\langle Bu_{\mathrm{z}\|}\rangle + \hat{\ell}_{11}^{\mathrm{eb}}\frac{n_\mathrm{e}}{n_\mathrm{b}}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle\\ &- \hat{\ell}_{12}^{\mathrm{ee}}n_\mathrm{e}\langle B\hat{q}_{\mathrm{e}\|}\rangle - \hat{\ell}_{12}^{\mathrm{ei}}n_\mathrm{e}\langle B\hat{q}_{\mathrm{i}\|}\rangle - \hat{\ell}_{12}^{\mathrm{ez}}n_\mathrm{e}\langle B\hat{q}_{\mathrm{z}\|}\rangle\\ &- e\frac{n_\mathrm{e}}{m_\mathrm{e}}I\langle R^{-2}\rangle\left(-\frac{\dot{\psi}_t}{q}+\dot{\psi}\right) - \nu_\mathrm{L}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle - \nu_{0\mathrm{e}}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle\\ &- \frac{\partial}{\partial V}\langle|\nabla\rho|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\langle Bu_{\mathrm{e}\|}\rangle + \langle|\nabla V|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\left(\frac{\partial}{\partial V}\frac{\langle B^2\rangle}{I}\right)\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\|}\rangle\\ &+ \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}n_\mathrm{e}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial V}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\|}\rangle\\ &- \langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}n_\mathrm{e}\left(\frac{\partial}{\partial V}\frac{\langle B^2\rangle}{I}\right)\frac{\partial}{\partial V}\frac{I}{\langle B^2\rangle}\langle Bu_{\mathrm{e}\|}\rangle - \frac{1}{m_\mathrm{e}}\frac{\langle B^2\rangle}{I}\frac{\partial}{\partial V}\langle\Pi_\mathrm{e}^{\mathrm{res}}\rangle\\ &- \frac{\partial}{\partial V}\left[\langle|\nabla V|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\frac{\langle R^2\rangle}{\langle B^2\rangle}\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{\partial\psi}{\partial V}(1+2\hat{q}^2)\frac{1}{I}\frac{\partial\psi}{\partial V}BV_{1\mathrm{e}}\right.\\ &\left.-\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}n_\mathrm{e}\frac{\partial}{\partial V}\frac{\langle R^2\rangle}{\langle B^2\rangle}\left\langle\frac{|\nabla V|^2}{R^2}\right\rangle\frac{\partial\psi}{\partial V}(1+2\hat{q}^2)\frac{1}{I}\frac{\partial\psi}{\partial V}BV_{1\mathrm{e}}\right]\end{aligned} \tag{3.8}$$

$$\begin{aligned}\frac{\partial}{\partial t}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle = &-(u_\mathrm{e}^V-u_g^V)\frac{\partial}{\partial V}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle - \frac{\partial u_\mathrm{e}^V}{\partial V}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle - \frac{\partial}{\partial V}[\langle|\nabla V|\rangle v_{\mathrm{e}\zeta}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle]\\ &+ \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}\frac{\partial}{\partial V}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle - \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\chi_{\mathrm{e}\zeta}\langle Ru_{\mathrm{e}\zeta}\rangle\frac{\partial}{\partial V}n_\mathrm{e} - \frac{1}{m_\mathrm{e}}\frac{\partial}{\partial V}\langle\Pi_\mathrm{e}^{\mathrm{res}}\rangle\\ &+ \hat{\ell}_{11}^{\mathrm{ee}}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle + \hat{\ell}_{11}^{\mathrm{ei}}\frac{n_\mathrm{e}}{n_\mathrm{i}}n_\mathrm{i}\langle Ru_{\mathrm{i}\zeta}\rangle + \hat{\ell}_{11}^{\mathrm{ez}}\frac{n_\mathrm{e}}{n_\mathrm{z}}n_\mathrm{z}\langle Ru_{\mathrm{z}\zeta}\rangle + \hat{\ell}_{11}^{\mathrm{eb}}\frac{n_\mathrm{e}}{n_\mathrm{b}}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle\end{aligned}$$

$$
\begin{aligned}
&- \hat{\ell}_{12}^{\mathrm{ee}} n_{\mathrm{e}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{e}} - \hat{\ell}_{12}^{\mathrm{ee}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle \\
&+ \hat{\ell}_{12}^{\mathrm{ei}} n_{\mathrm{e}} \frac{1}{Z_{\mathrm{i}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{i}} - \hat{\ell}_{12}^{\mathrm{ei}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle \\
&+ \hat{\ell}_{12}^{\mathrm{ez}} n_{\mathrm{e}} \frac{1}{Z_{\mathrm{z}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{z}} - \hat{\ell}_{12}^{\mathrm{ez}} n_{\mathrm{e}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
&- e \frac{n_{\mathrm{e}}}{m_{\mathrm{e}}} \dot{\psi} - e \frac{1}{m_{\mathrm{e}}} \frac{\partial \psi}{\partial V} n_{\mathrm{e}} u_{\mathrm{e}}^V \\
&- \frac{e^2 \langle |\nabla V|^2 \rangle D_{\mathrm{e}}}{m_{\mathrm{e}} T_{\mathrm{e}}} \frac{\partial \psi}{\partial V} \left[n_{\mathrm{e}} \frac{\partial \psi}{\partial V} \frac{1}{I} B V_{1\mathrm{e}} + n_{\mathrm{e}} \frac{\partial}{\partial V} \Phi + \frac{C_{\mathrm{T}} - 1}{e} \left(\frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - T_{\mathrm{e}} \frac{\partial}{\partial V} n_{\mathrm{e}} \right) + \frac{T_{\mathrm{e}}}{e} \frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{e}} \right] \\
&- \nu_{\mathrm{L}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle - \nu_{0\mathrm{e}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle
\end{aligned} \tag{3.9}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{\partial}{\partial t} n_{\mathrm{e}} T_{\mathrm{e}} = &- \frac{5}{2} (u_{\mathrm{e}}^V - u_g^V) \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{5}{2} \frac{\partial u_{\mathrm{e}}^V}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{\partial}{\partial V} [V_{\mathrm{e}}^{\mathrm{hp}} \langle |\nabla V| \rangle n_{\mathrm{e}} T_{\mathrm{e}}] \\
&+ \frac{\partial}{\partial V} \chi_{\mathrm{e}} \langle |\nabla V|^2 \rangle \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{\partial}{\partial V} \chi_{\mathrm{e}} \langle |\nabla V|^2 \rangle T_{\mathrm{e}} \frac{\partial}{\partial V} n_{\mathrm{e}} \\
&- u_g^V \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - u_{\mathrm{i}}^V \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - u_{\mathrm{z}}^V \frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \hat{u}_{\mathrm{i}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{i}} \rangle - \hat{u}_{\mathrm{z}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{z}} \rangle \\
&- \frac{3}{2} \nu_{\mathrm{Tei}} \left(n_{\mathrm{e}} T_{\mathrm{e}} - \frac{n_{\mathrm{e}}}{n_{\mathrm{i}}} n_{\mathrm{i}} T_{\mathrm{i}} \right) - \frac{3}{2} \nu_{\mathrm{Tez}} \left(n_{\mathrm{e}} T_{\mathrm{e}} - \frac{n_{\mathrm{e}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} \right) \\
&- 4\pi^2 \frac{\partial \psi}{\partial V} \dot{\psi}_t (-e n_{\mathrm{e}} \hat{u}_{\mathrm{e}\theta} + Z_{\mathrm{i}} e n_{\mathrm{i}} \hat{u}_{\mathrm{i}\theta} + Z_{\mathrm{z}} e n_{\mathrm{z}} \hat{u}_{\mathrm{z}\theta}) \\
&+ \dot{\psi} \left(-e n_{\mathrm{e}} \left\langle \frac{u_{\mathrm{e}\zeta}}{R} \right\rangle + Z_{\mathrm{i}} e n_{\mathrm{i}} \left\langle \frac{u_{\mathrm{i}\zeta}}{R} \right\rangle + Z_{\mathrm{z}} e n_{\mathrm{z}} \left\langle \frac{u_{\mathrm{z}\zeta}}{R} \right\rangle \right) \\
&- \nu_{\mathrm{L}} (n_{\mathrm{e}} T_{\mathrm{e}} - n_{\mathrm{esup}} T_{\mathrm{e}}) - \frac{3}{2} \nu_{\mathrm{LT_e}} (n_{\mathrm{e}} T_{\mathrm{e}} - n_{\mathrm{e}} T_{\mathrm{esup}}) - E_{\mathrm{H}} \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) \\
&+ f_{\mathrm{ei}} P_{\mathrm{b}} + P_{\alpha \mathrm{e}} + P_{\mathrm{RFe}} - P_{\mathrm{Br}}
\end{aligned} \tag{3.10}
$$

$$
\begin{aligned}
\frac{5}{2} n_{\mathrm{e}} \frac{\partial}{\partial t} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle = &- \frac{1}{m_{\mathrm{e}}} \hat{\mu}_2^{\mathrm{e}} \langle B u_{\mathrm{e}\parallel} \rangle - \frac{1}{m_{\mathrm{e}}} \hat{\mu}_3^{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \frac{1}{m_{\mathrm{e}}} \hat{\mu}_2^{\mathrm{e}} B V_{1\mathrm{e}} \\
&+ \frac{1}{e m_{\mathrm{e}}} \hat{\mu}_3^{\mathrm{e}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{e}}} \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - \frac{1}{e m_{\mathrm{e}}} \hat{\mu}_3^{\mathrm{e}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{e}}}{n_{\mathrm{e}}} \frac{\partial}{\partial V} n_{\mathrm{e}} \\
&- \hat{\ell}_{21}^{\mathrm{ee}} n_{\mathrm{e}} \langle B u_{\mathrm{e}\parallel} \rangle - \hat{\ell}_{21}^{\mathrm{ei}} n_{\mathrm{e}} \langle B u_{\mathrm{i}\parallel} \rangle - \hat{\ell}_{21}^{\mathrm{ez}} n_{\mathrm{e}} \langle B u_{\mathrm{z}\parallel} \rangle - \hat{\ell}_{21}^{\mathrm{eb}} \frac{n_{\mathrm{e}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\
&+ \hat{\ell}_{22}^{\mathrm{ee}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle + \hat{\ell}_{22}^{\mathrm{ei}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \hat{\ell}_{22}^{\mathrm{ez}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
&- \nu_{\mathrm{L}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle - \nu_{0\mathrm{e}} n_{\mathrm{e}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle
\end{aligned} \tag{3.11}
$$

$$
0 = - n_{\mathrm{e}} \left\langle \frac{u_{\mathrm{e}\zeta}}{R} \right\rangle + \frac{1}{\langle R^2 \rangle} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle - \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} n_{\mathrm{e}} \langle B u_{\mathrm{e}\parallel} \rangle \tag{3.12}
$$

$$0 = -n_{\mathrm{e}} B V_{1\mathrm{e}} + \frac{1}{e}\frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - n_{\mathrm{e}} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} \Phi \tag{3.13}$$

$$\begin{aligned}\frac{\partial}{\partial t} n_{\mathrm{i}} = &-\frac{\partial}{\partial V} n_{\mathrm{i}} u_{\mathrm{i}}^{V} + u_g^{V} \frac{\partial}{\partial V} n_{\mathrm{i}} + \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} + n_{02} + n_{03}) - \nu_{\mathrm{L}} (n_{\mathrm{i}} - n_{\mathrm{isup}}) \\ &+ \nu_{\mathrm{b}} n_{\mathrm{b}} + \nu_{\mathrm{b}} g n_{\mathrm{b}}^{\mathrm{rp}} - f_{\mathrm{CX}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} + \nu_{\mathrm{bL}} n_{\mathrm{b}} + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{i}}\end{aligned} \tag{3.14}$$

$$0 = Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} n_{\mathrm{i}} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} B V_{1\mathrm{i}} + Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} I n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} \langle B^2 \rangle n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle \tag{3.15}$$

$$\begin{aligned}n_{\mathrm{i}} \frac{\partial}{\partial t} \langle B u_{\mathrm{i}\parallel} \rangle = &-\frac{1}{m_{\mathrm{i}}} \hat{\mu}_1^{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle + \frac{1}{m_{\mathrm{i}}} \hat{\mu}_1^{\mathrm{i}} B V_{1\mathrm{i}} \\ &- \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} + \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{i}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{i}}}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} \\ &+ \hat{\ell}_{11}^{\mathrm{ie}} n_{\mathrm{i}} \langle B u_{\mathrm{e}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{ii}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{iz}} n_{\mathrm{i}} \langle B u_{\mathrm{z}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{ib}} \frac{n_{\mathrm{i}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\ &- \hat{\ell}_{12}^{\mathrm{ie}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{ii}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{iz}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\ &+ Z_{\mathrm{i}} e \frac{n_{\mathrm{i}}}{m_{\mathrm{i}}} I \langle R^{-2} \rangle \left(-\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) \\ &- \nu_{\mathrm{L}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle + \nu_{\mathrm{b}} \frac{m_{\mathrm{b}}}{m_{\mathrm{i}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle - \nu_{0\mathrm{i}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \nu_{\mathrm{CX}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \nu_{\mathrm{LC}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle - \nu_{\mathrm{OL}} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle \\ &- \frac{\partial}{\partial V} \langle |\nabla \rho| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \langle B u_{\mathrm{i}\parallel} \rangle + \langle |\nabla V| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{i}\parallel} \rangle \\ &+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} n_{\mathrm{i}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{i}\parallel} \rangle \\ &- \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} n_{\mathrm{i}} \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{i}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{i}}^{\mathrm{res}} \rangle \\ &- \frac{\partial}{\partial V} \left[\langle |\nabla V| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{i}} \right. \\ &\left. - \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} n_{\mathrm{i}} \frac{\partial}{\partial V} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{i}} \right]\end{aligned} \tag{3.16}$$

$$\begin{aligned}\frac{\partial}{\partial t} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle = &-(u_{\mathrm{i}}^{V} - u_g^{V}) \frac{\partial}{\partial V} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{\partial u_{\mathrm{i}}^{V}}{\partial V} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V| \rangle v_{\mathrm{i}\zeta} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle \\ &+ \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} \frac{\partial}{\partial V} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{i}\zeta} \langle R u_{\mathrm{i}\zeta} \rangle \frac{\partial}{\partial V} n_{\mathrm{i}} - \frac{1}{m_{\mathrm{i}}} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{i}}^{\mathrm{res}} \rangle \\ &+ \hat{\ell}_{11}^{\mathrm{ie}} \frac{n_{\mathrm{i}}}{n_{\mathrm{e}}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{ii}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{iz}} \frac{n_{\mathrm{i}}}{n_{\mathrm{z}}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{ib}} \frac{n_{\mathrm{i}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle\end{aligned}$$

$$
\begin{aligned}
&- \hat{\ell}^{\mathrm{ie}}_{12} n_{\mathrm{i}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{e}} - \hat{\ell}^{\mathrm{ie}}_{12} n_{\mathrm{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{e}\|} \rangle \\
&+ \hat{\ell}^{\mathrm{ii}}_{12} n_{\mathrm{i}} \frac{1}{Z_{\mathrm{i}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{i}} - \hat{\ell}^{\mathrm{ii}}_{12} n_{\mathrm{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{i}\|} \rangle \\
&+ \hat{\ell}^{\mathrm{iz}}_{12} n_{\mathrm{i}} \frac{1}{Z_{\mathrm{z}} e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{z}} - \hat{\ell}^{\mathrm{iz}}_{12} n_{\mathrm{i}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{z}\|} \rangle \\
&+ Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} n_{\mathrm{i}} \dot{\psi} + Z_{\mathrm{i}} \frac{e}{m_{\mathrm{i}}} \frac{\partial \psi}{\partial V} n_{\mathrm{i}} u^V_{\mathrm{i}} \\
&+ Z_{\mathrm{i}} \frac{e^2 \langle |\nabla V|^2 \rangle D_{\mathrm{e}}}{m_{\mathrm{i}} T_{\mathrm{e}}} \frac{\partial \psi}{\partial V} \left[(n_{\mathrm{i}} + n_{\mathrm{b}}) \frac{\partial \psi}{\partial V} \frac{1}{I} B V_{1\mathrm{e}} + (n_{\mathrm{i}} + n_{\mathrm{b}}) \frac{\partial}{\partial V} \Phi \right. \\
&\left. + \frac{C_{\mathrm{T}} - 1}{e} \frac{n_{\mathrm{i}} + n_{\mathrm{b}}}{n_{\mathrm{e}}} \left(\frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - T_{\mathrm{e}} \frac{\partial}{\partial V} n_{\mathrm{e}} \right) + \frac{T_{\mathrm{e}}}{e} \frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{i}} + \frac{T_{\mathrm{e}}}{e} \frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{b}} \right] \\
&- \nu_{\mathrm{L}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \nu_{\mathrm{b}} \frac{m_{\mathrm{b}}}{m_{\mathrm{i}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\zeta} \rangle - \nu_{0\mathrm{i}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \nu_{\mathrm{CX}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \nu_{\mathrm{OL}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle
\end{aligned} \tag{3.17}
$$

$$
\begin{aligned}
\frac{3}{2} \frac{\partial}{\partial t} n_{\mathrm{i}} T_{\mathrm{i}} = &- \frac{5}{2} (u^V_{\mathrm{i}} - u^V_g) \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - \frac{5}{2} \frac{\partial u^V_{\mathrm{i}}}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - \frac{\partial}{\partial V} V^{\mathrm{hp}}_{\mathrm{i}} \langle |\nabla V| \rangle n_{\mathrm{i}} T_{\mathrm{i}} \\
&+ \frac{\partial}{\partial V} \chi_{\mathrm{i}} \langle |\nabla V|^2 \rangle \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - \frac{\partial}{\partial V} \chi_{\mathrm{i}} \langle |\nabla V|^2 \rangle T_{\mathrm{i}} \frac{\partial}{\partial V} n_{\mathrm{i}} \\
&+ (u^V_{\mathrm{i}} - u^V_g) \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} + \hat{u}_{\mathrm{i}\theta} \langle \boldsymbol{B} \cdot \nabla \cdot \overleftrightarrow{\pi}_{\mathrm{i}} \rangle - \frac{3}{2} \nu_{\mathrm{Tei}} \left(\frac{n_{\mathrm{e}}}{n_{\mathrm{i}}} n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{e}} T_{\mathrm{e}} \right) - \frac{3}{2} \nu_{\mathrm{Tiz}} \left(n_{\mathrm{i}} T_{\mathrm{i}} - \frac{n_{\mathrm{i}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} \right) \\
&- \nu_{\mathrm{L}} \left(n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{isup}} T_{\mathrm{i}} \right) - \frac{3}{2} \nu_{\mathrm{LT_i}} (n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{i}} T_{\mathrm{isup}}) \\
&+ \frac{3}{2} \nu_{\mathrm{ionize}} \frac{n_{\mathrm{e}}}{n_{01} + n_{02} + n_{03}} (n_{01} T_{01} + n_{02} T_{02} + n_{03} T_{03}) \\
&- \frac{3}{2} \nu_{\mathrm{CX}} (n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{i}} T_{01}) + (1 - f_{\mathrm{ei}}) P_{\mathrm{b}} + m_{\mathrm{b}} \frac{1}{n_{\mathrm{b}}} \frac{P_{\mathrm{b}}}{E_{\mathrm{b}}} \frac{\langle B u_{\mathrm{i}\|} \rangle}{\langle B^2 \rangle} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle + P_{\alpha \mathrm{i}} + P_{\mathrm{RFi}}
\end{aligned} \tag{3.18}
$$

$$
\begin{aligned}
\frac{5}{2} n_{\mathrm{i}} \frac{\partial}{\partial t} \langle B \hat{q}_{\mathrm{i}\|} \rangle = &- \frac{1}{m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_2 \langle B u_{\mathrm{i}\|} \rangle - \frac{1}{m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_3 \langle B \hat{q}_{\mathrm{i}\|} \rangle + \frac{1}{m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_2 B V_{1\mathrm{i}} \\
&- \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_3 I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} + \frac{1}{Z_{\mathrm{i}} e m_{\mathrm{i}}} \hat{\mu}^{\mathrm{i}}_3 I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{i}}}{n_{\mathrm{i}}} \frac{\partial}{\partial V} n_{\mathrm{i}} \\
&- \hat{\ell}^{\mathrm{ie}}_{21} n_{\mathrm{i}} \langle B u_{\mathrm{e}\|} \rangle - \hat{\ell}^{\mathrm{ii}}_{21} n_{\mathrm{i}} \langle B u_{\mathrm{i}\|} \rangle - \hat{\ell}^{\mathrm{iz}}_{21} n_{\mathrm{i}} \langle B u_{\mathrm{z}\|} \rangle - \hat{\ell}^{\mathrm{ib}}_{21} \frac{n_{\mathrm{i}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\|} \rangle \\
&+ \hat{\ell}^{\mathrm{ie}}_{22} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{e}\|} \rangle + \hat{\ell}^{\mathrm{ii}}_{22} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\|} \rangle + \hat{\ell}^{\mathrm{iz}}_{22} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{z}\|} \rangle \\
&- \nu_{\mathrm{L}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\|} \rangle - \nu_{0\mathrm{i}} n_{\mathrm{i}} \langle B \hat{q}_{\mathrm{i}\|} \rangle
\end{aligned} \tag{3.19}
$$

$$
0 = - n_{\mathrm{i}} \left\langle \frac{u_{\mathrm{i}\zeta}}{R} \right\rangle + \frac{1}{\langle R^2 \rangle} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle - \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{\langle R^2 \rangle} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \frac{2\hat{q}^2}{1 + 2\hat{q}^2} \frac{1}{I} n_{\mathrm{i}} \langle B u_{\mathrm{i}\|} \rangle \tag{3.20}
$$

$$0 = -n_{\mathrm{i}} B V_{1\mathrm{i}} - \frac{1}{Z_{\mathrm{i}} e} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} n_{\mathrm{i}} T_{\mathrm{i}} - n_{\mathrm{i}} \frac{\partial V}{\partial \psi} I \frac{\partial}{\partial V} \Phi \tag{3.21}$$

$$\frac{\partial}{\partial t} n_{\mathrm{z}} = -\frac{\partial}{\partial V} n_{\mathrm{z}} u_{\mathrm{z}}^{V} + u_g^{V} \frac{\partial}{\partial V} n_{\mathrm{z}} + n_{\mathrm{e}} \bar{n}_z f_5 S_5 - n_{\mathrm{e}} n_{\mathrm{z}} A_6 - \nu_{\mathrm{L}} (n_{\mathrm{z}} - n_{\mathrm{isup}}) + S_{\mathrm{LC}} - \nu_{\mathrm{OL}} n_{\mathrm{z}} \tag{3.22}$$

$$0 = Z_{\mathrm{z}} \frac{e}{m_{\mathrm{z}}} n_{\mathrm{z}} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{1}{I} B V_{1\mathrm{z}} + Z_{\mathrm{z}} \frac{e}{m_{\mathrm{z}}} I n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - Z_{\mathrm{z}} \frac{e}{m_{\mathrm{z}}} \langle B^2 \rangle n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \tag{3.23}$$

$$\begin{aligned}
n_{\mathrm{z}} \frac{\partial}{\partial t} \langle B u_{\mathrm{z}\parallel} \rangle = & -\frac{1}{m_{\mathrm{i}}} \hat{\mu}_1^{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \frac{1}{m_{\mathrm{i}}} \hat{\mu}_2^{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle + \frac{1}{m_{\mathrm{z}}} \hat{\mu}_1^{\mathrm{i}} B V_{1\mathrm{z}} \\
& - \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}} \hat{\mu}_2^{\mathrm{z}} I \frac{\partial V}{\partial \psi} \frac{1}{n_{\mathrm{z}}} \frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} + \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}} \hat{\mu}_2^{\mathrm{z}} I \frac{\partial V}{\partial \psi} \frac{T_{\mathrm{z}}}{n_{\mathrm{z}}} \frac{\partial}{\partial V} n_{\mathrm{z}} \\
& + \hat{\ell}_{11}^{\mathrm{ze}} n_{\mathrm{z}} \langle B u_{\mathrm{e}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{zi}} n_{\mathrm{z}} \langle B u_{\mathrm{i}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{zz}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle + \hat{\ell}_{11}^{\mathrm{zb}} \frac{n_{\mathrm{z}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle B u_{\mathrm{b}\parallel} \rangle \\
& - \hat{\ell}_{12}^{\mathrm{ze}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{zi}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{i}\parallel} \rangle - \hat{\ell}_{12}^{\mathrm{zz}} n_{\mathrm{z}} \langle B \hat{q}_{\mathrm{z}\parallel} \rangle \\
& + Z_{\mathrm{z}} e \frac{n_{\mathrm{z}}}{m_{\mathrm{z}}} I \langle R^{-2} \rangle \left(-\frac{\dot{\psi}_t}{q} + \dot{\psi} \right) \\
& - \nu_{\mathrm{L}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{0\mathrm{z}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle - \nu_{\mathrm{OL}} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle \\
& - \frac{\partial}{\partial V} \langle |\nabla \rho| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \langle B u_{\mathrm{z}\parallel} \rangle + \langle |\nabla V| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
& + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} n_{\mathrm{z}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle \\
& - \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} n_{\mathrm{z}} \left(\frac{\partial}{\partial V} \frac{\langle B^2 \rangle}{I} \right) \frac{\partial}{\partial V} \frac{I}{\langle B^2 \rangle} \langle B u_{\mathrm{z}\parallel} \rangle - \frac{1}{m_{\mathrm{z}}} \frac{\langle B^2 \rangle}{I} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \\
& - \frac{\partial}{\partial V} \left[\langle |\nabla V| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{z}} \right. \\
& \left. - \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} n_{\mathrm{z}} \frac{\partial}{\partial V} \frac{\langle R^2 \rangle}{\langle B^2 \rangle} \left\langle \frac{|\nabla V|^2}{R^2} \right\rangle \frac{\partial \psi}{\partial V} (1 + 2\hat{q}^2) \frac{1}{I} \frac{\partial \psi}{\partial V} B V_{1\mathrm{z}} \right]
\end{aligned} \tag{3.24}$$

$$\begin{aligned}
\frac{\partial}{\partial t} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle = & -(u_{\mathrm{z}}^{V} - u_g^{V}) \frac{\partial}{\partial V} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \frac{\partial u_{\mathrm{z}}^{V}}{\partial V} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V| \rangle v_{\mathrm{z}\zeta} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle \\
& + \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} \frac{\partial}{\partial V} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle - \frac{\partial}{\partial V} \langle |\nabla V|^2 \rangle \chi_{\mathrm{z}\zeta} \langle R u_{\mathrm{z}\zeta} \rangle \frac{\partial}{\partial V} n_{\mathrm{z}} - \frac{1}{m_{\mathrm{z}}} \frac{\partial}{\partial V} \langle \Pi_{\mathrm{z}}^{\mathrm{res}} \rangle \\
& + \hat{\ell}_{11}^{\mathrm{ze}} \frac{n_{\mathrm{z}}}{n_{\mathrm{e}}} n_{\mathrm{e}} \langle R u_{\mathrm{e}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{zi}} \frac{n_{\mathrm{z}}}{n_{\mathrm{i}}} n_{\mathrm{i}} \langle R u_{\mathrm{i}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{zz}} n_{\mathrm{z}} \langle R u_{\mathrm{z}\zeta} \rangle + \hat{\ell}_{11}^{\mathrm{zb}} \frac{n_{\mathrm{z}}}{n_{\mathrm{b}}} n_{\mathrm{b}} \langle R u_{\mathrm{b}\zeta} \rangle \\
& - \hat{\ell}_{12}^{\mathrm{ze}} n_{\mathrm{z}} \frac{1}{e} \frac{1}{\langle B^2 \rangle} \left(\langle B^2 \rangle \langle R^2 \rangle - I^2 \right) \frac{\partial V}{\partial \psi} \frac{\partial}{\partial V} T_{\mathrm{e}} - \hat{\ell}_{12}^{\mathrm{ze}} n_{\mathrm{z}} \frac{I}{\langle B^2 \rangle} \langle B \hat{q}_{\mathrm{e}\parallel} \rangle
\end{aligned}$$

$$
\begin{aligned}
&+\hat{\ell}_{12}^{\mathrm{zi}} n_{\mathrm{z}} \frac{1}{Z_{\mathrm{i}} e} \frac{1}{\langle B^2\rangle}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{\partial V}{\partial \psi}\frac{\partial}{\partial V}T_{\mathrm{i}} - \hat{\ell}_{12}^{\mathrm{zi}} n_{\mathrm{z}} \frac{I}{\langle B^2\rangle}\langle B\hat{q}_{\mathrm{i}\|}\rangle \\
&+\hat{\ell}_{12}^{\mathrm{zz}} n_{\mathrm{z}} \frac{1}{Z_{\mathrm{z}} e} \frac{1}{\langle B^2\rangle}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{\partial V}{\partial \psi}\frac{\partial}{\partial V}T_{\mathrm{z}} - \hat{\ell}_{12}^{\mathrm{zz}} n_{\mathrm{z}} \frac{I}{\langle B^2\rangle}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&+ Z_{\mathrm{z}}\frac{e}{m_{\mathrm{z}}} n_{\mathrm{z}}\dot{\psi} + Z_{\mathrm{z}}\frac{e}{m_{\mathrm{z}}}\frac{\partial \psi}{\partial V} n_{\mathrm{z}} u_{\mathrm{z}}^{V} \\
&+ Z_{\mathrm{z}}\frac{e\langle|\nabla V|^2\rangle D_{\mathrm{e}}}{m_{\mathrm{z}} T_{\mathrm{e}}}\frac{\partial \psi}{\partial V}\left[n_{\mathrm{z}}\frac{\partial \psi}{\partial V}\frac{1}{I} B V_{1\mathrm{e}} + n_{\mathrm{z}}\frac{\partial}{\partial V}\Phi\right. \\
&\left.+\frac{C_{\mathrm{T}}-1}{e}\frac{n_{\mathrm{z}}}{n_{\mathrm{e}}}\left(\frac{\partial}{\partial V} n_{\mathrm{e}} T_{\mathrm{e}} - T_{\mathrm{e}}\frac{\partial}{\partial V} n_{\mathrm{e}}\right) + \frac{T_{\mathrm{e}}}{e}\frac{\partial \psi}{\partial V} C_{\mathrm{p}} n_{\mathrm{z}}\right] \\
&- \nu_{\mathrm{L}} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle - \nu_{0\mathrm{z}} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle - \nu_{\mathrm{OL}} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle
\end{aligned} \tag{3.25}
$$

$$
\begin{aligned}
\frac{3}{2}\frac{\partial}{\partial t} n_{\mathrm{z}} T_{\mathrm{z}} &= -\frac{5}{2}(u_{\mathrm{z}}^{V} - u_{g}^{V})\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \frac{5}{2}\frac{\partial u_{\mathrm{z}}^{V}}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \frac{\partial}{\partial V} V_{\mathrm{z}}^{\mathrm{hp}}\langle|\nabla V|\rangle n_{\mathrm{z}} T_{\mathrm{z}} \\
&+ \frac{\partial}{\partial V}\chi_{\mathrm{z}}\langle|\nabla V|^2\rangle\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - \frac{\partial}{\partial V}\chi_{\mathrm{z}}\langle|\nabla V|^2\rangle T_{\mathrm{z}}\frac{\partial}{\partial V} n_{\mathrm{z}} \\
&+ (u_{\mathrm{z}}^{V} - u_{g}^{V})\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} + \hat{u}_{\mathrm{z}\theta}\langle \boldsymbol{B}\cdot\nabla\cdot\overleftrightarrow{\pi}_{\mathrm{z}}\rangle - \frac{3}{2}\nu_{\mathrm{Tez}}\left(\frac{n_{\mathrm{e}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{e}} T_{\mathrm{e}}\right) - \frac{3}{2}\nu_{\mathrm{Tiz}}\left(\frac{n_{\mathrm{i}}}{n_{\mathrm{z}}} n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{i}} T_{\mathrm{i}}\right) \\
&- \nu_{\mathrm{L}}\left(n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{zsup}} T_{\mathrm{z}}\right) - \frac{3}{2}\nu_{\mathrm{LT_z}}\left(n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{z}} T_{\mathrm{zsup}}\right) + P_{\alpha\mathrm{z}} + P_{\mathrm{RFz}}
\end{aligned} \tag{3.26}
$$

$$
\begin{aligned}
\frac{5}{2} n_{\mathrm{z}}\frac{\partial}{\partial t}\langle B\hat{q}_{\mathrm{z}\|}\rangle &= -\frac{1}{m_{\mathrm{z}}}\hat{\mu}_2^{\mathrm{z}}\langle B u_{\mathrm{z}\|}\rangle - \frac{1}{m_{\mathrm{z}}}\hat{\mu}_3^{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle + \frac{1}{m_{\mathrm{z}}}\hat{\mu}_2^{\mathrm{z}} B V_{1\mathrm{z}} \\
&- \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}}\hat{\mu}_3^{\mathrm{z}} I\frac{\partial V}{\partial \psi}\frac{1}{n_{\mathrm{z}}}\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} + \frac{1}{Z_{\mathrm{z}} e m_{\mathrm{z}}}\hat{\mu}_3^{\mathrm{z}} I\frac{\partial V}{\partial \psi}\frac{T_{\mathrm{z}}}{n_{\mathrm{z}}}\frac{\partial}{\partial V} n_{\mathrm{z}} \\
&- \hat{\ell}_{21}^{\mathrm{ze}} n_{\mathrm{z}}\langle B u_{\mathrm{e}\|}\rangle - \hat{\ell}_{21}^{\mathrm{zi}} n_{\mathrm{z}}\langle B u_{\mathrm{i}\|}\rangle - \hat{\ell}_{21}^{\mathrm{zz}} n_{\mathrm{z}}\langle B u_{\mathrm{z}\|}\rangle - \hat{\ell}_{21}^{\mathrm{zb}}\frac{n_{\mathrm{z}}}{n_{\mathrm{b}}} n_{\mathrm{b}}\langle B u_{\mathrm{b}\|}\rangle \\
&+ \hat{\ell}_{22}^{\mathrm{ze}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{e}\|}\rangle + \hat{\ell}_{22}^{\mathrm{zi}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{i}\|}\rangle + \hat{\ell}_{22}^{\mathrm{zz}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle \\
&- \nu_{\mathrm{L}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle - \nu_{0\mathrm{z}} n_{\mathrm{z}}\langle B\hat{q}_{\mathrm{z}\|}\rangle
\end{aligned} \tag{3.27}
$$

$$
0 = -n_{\mathrm{z}}\left\langle\frac{u_{\mathrm{z}\zeta}}{R}\right\rangle + \frac{1}{\langle R^2\rangle} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle - \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{\langle R^2\rangle} n_{\mathrm{z}}\langle R u_{\mathrm{z}\zeta}\rangle + \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{I} n_{\mathrm{z}}\langle B u_{\mathrm{z}\|}\rangle \tag{3.28}
$$

$$
0 = -n_{\mathrm{z}} B V_{1\mathrm{z}} - \frac{1}{Z_{\mathrm{z}} e}\frac{\partial V}{\partial \psi} I\frac{\partial}{\partial V} n_{\mathrm{z}} T_{\mathrm{z}} - n_{\mathrm{z}}\frac{\partial V}{\partial \psi} I\frac{\partial}{\partial V}\Phi \tag{3.29}
$$

$$\frac{\partial}{\partial t}n_\mathrm{b} = -\frac{\partial}{\partial V}n_\mathrm{b}u_\mathrm{b}^V + u_g^V\frac{\partial}{\partial V}n_\mathrm{b} + \frac{P_\mathrm{b}}{E_\mathrm{b}} - g\frac{P_{\mathrm{b}\perp}}{E_\mathrm{b}} - \nu_\mathrm{b}n_\mathrm{b} - \nu_\mathrm{bL}n_\mathrm{b} - \nu_\mathrm{eff}g(\sqrt{\delta}n_\mathrm{b} - n_\mathrm{b}^\mathrm{rp})$$
$$+ \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle f_t D_\mathrm{b}\frac{\partial}{\partial V}n_\mathrm{b} \tag{3.30}$$

$$0 = Z_\mathrm{b}\frac{e}{m_\mathrm{b}}\left(\langle B^2\rangle\langle R^2\rangle - I^2\right)\frac{1}{I}n_\mathrm{b}BV_{1\mathrm{b}} + Z_\mathrm{b}\frac{e}{m_\mathrm{b}}In_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle - Z_\mathrm{b}\frac{e}{m_\mathrm{b}}\langle B^2\rangle n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle \tag{3.31}$$

$$\frac{\partial}{\partial t}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle = -\frac{1}{m_\mathrm{b}n_\mathrm{b}}\hat{\mu}_1^\mathrm{b}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle + \frac{1}{m_\mathrm{b}n_\mathrm{b}}\hat{\mu}_1^\mathrm{b}n_\mathrm{b}BV_{1\mathrm{b}}$$
$$+ \hat{\ell}_{11}^{\mathrm{bb}}\frac{m_\mathrm{e}n_\mathrm{e}}{m_\mathrm{b}n_\mathrm{b}}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle + \sum_{a\neq\mathrm{b}}\hat{\ell}_{11}^{\mathrm{b}a}\frac{m_\mathrm{e}n_\mathrm{e}}{m_\mathrm{b}}\langle Bu_{a\|}\rangle$$
$$+ Z_\mathrm{b}\frac{e}{m_\mathrm{b}}n_\mathrm{b}I\langle R^{-2}\rangle\left(-\frac{\dot{\psi}_t}{q} + \dot{\psi}\right) - \nu_\mathrm{bL}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle - \nu_\mathrm{b}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle - \nu_{0\mathrm{b}}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle$$
$$- \nu_\mathrm{CX}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle + \frac{1}{m_\mathrm{b}}\langle BS_{\mathrm{m}\|\mathrm{b}}\rangle - \nu_\mathrm{eff}g\sqrt{\delta}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle\langle Bu_{\mathrm{b}\|}\rangle f_t D_\mathrm{b}\frac{\partial}{\partial V}n_\mathrm{b} \tag{3.32}$$

$$\frac{\partial}{\partial t}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle = u_g^V\frac{\partial}{\partial V}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \frac{\partial}{\partial V}\langle Ru_{\mathrm{b}\zeta}\rangle n_\mathrm{b}u_\mathrm{b}^V$$
$$+ \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{be}}\frac{1}{n_\mathrm{e}}n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{bi}}\frac{1}{n_\mathrm{i}}n_\mathrm{i}\langle Ru_{\mathrm{i}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{bz}}\frac{1}{n_\mathrm{z}}n_\mathrm{z}\langle Ru_{\mathrm{z}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\ell_{11}^{\mathrm{bb}}\frac{1}{n_\mathrm{b}}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle$$
$$+ Z_\mathrm{b}\frac{e}{m_\mathrm{b}}n_\mathrm{b}\dot{\psi} + Z_\mathrm{b}\frac{e}{m_\mathrm{b}}\frac{\partial\psi}{\partial V}n_\mathrm{b}u_\mathrm{b}^V$$
$$- \nu_\mathrm{bL}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \nu_\mathrm{b}n_\mathrm{b}\langle Bu_{\mathrm{b}\zeta}\rangle - \nu_{0\mathrm{i}}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \nu_\mathrm{CX}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle + \frac{1}{m_\mathrm{b}}\frac{I}{\langle B^2\rangle}\langle BS_{\mathrm{m}\|\mathrm{b}}\rangle \tag{3.33}$$

$$0 = -n_\mathrm{b}\left\langle\frac{u_{\mathrm{b}\zeta}}{R}\right\rangle + \frac{1}{\langle R^2\rangle}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle - \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{\langle R^2\rangle}n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle + \frac{2\hat{q}^2}{1+2\hat{q}^2}\frac{1}{I}n_\mathrm{b}\langle Bu_{\mathrm{b}\|}\rangle \tag{3.34}$$

$$0 = -n_\mathrm{b}BV_{1\mathrm{b}} - \frac{1}{Z_\mathrm{b}e}\frac{\partial V}{\partial\psi}In_\mathrm{b}\frac{\partial}{\partial V}T_\mathrm{b} - \frac{1}{Z_\mathrm{b}e}\frac{\partial V}{\partial\psi}IT_\mathrm{b}\frac{\partial}{\partial V}n_\mathrm{b} - n_\mathrm{b}\frac{\partial V}{\partial\psi}I\frac{\partial}{\partial V}\Phi \tag{3.35}$$

$$\frac{\partial}{\partial t}n_{01} = -\frac{\partial u_g^V}{\partial V}n_{01} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{01}\frac{\partial}{\partial V}n_{01} - \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}n_{01}$$
$$- \nu_\mathrm{CX}\frac{n_\mathrm{i}}{n_{01}+n_{02}}n_{01} + \gamma_0\nu_\mathrm{L}(n_\mathrm{i} - n_\mathrm{isup}) + \frac{\partial}{\partial V}V'\Gamma^\mathrm{puff}\Big|_\mathrm{b} \tag{3.36}$$

$$\frac{\partial}{\partial t}n_{02} = -\frac{\partial u_g^V}{\partial V}n_{02} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{02}\frac{\partial}{\partial V}n_{02} - \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}n_{02} + \nu_\mathrm{CX}\frac{n_\mathrm{i}}{n_{01}+n_{02}}n_{01} \tag{3.37}$$

$$\frac{\partial}{\partial t}n_{03} = -\frac{\partial u_g^V}{\partial V}n_{03} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{03}\frac{\partial}{\partial V}n_{03} - \nu_\mathrm{ionize}\frac{n_\mathrm{e}}{n_{01}+n_{02}+n_{03}}n_{03} + f_\mathrm{CX}\frac{P_\mathrm{b}}{E_\mathrm{b}} \tag{3.38}$$

$$\frac{\partial}{\partial t}\bar{n}_z = -\frac{\partial u_g^V}{\partial V}\bar{n}_z + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_{0z}\frac{\partial}{\partial V}\bar{n}_z - n_e\bar{n}_z f_5 S_5 + n_e n_z A_6$$
$$+ \gamma_{0z}\nu_L\left(n_z - n_{zsup}\right) + \gamma_{i,z}\nu_L\left(n_i - n_{isup}\right) + \frac{\partial}{\partial V}V'\Gamma_z^{puff}\Big|_b \tag{3.39}$$

$$\frac{\partial}{\partial t}n_b^{rp} = -\frac{\partial u_g^V}{\partial V}n_b^{rp}\frac{P_{b\perp}}{E_b} + \nu_{eff}(\sqrt{\delta}n_b - n_b^{rp}) - \nu_b n_b^{rp}$$
$$- \frac{\partial}{\partial V}\langle|\nabla V|\rangle u_b^{rp} n_b^{rp} + \frac{\partial}{\partial V}\langle|\nabla V|^2\rangle D_b^{rp}\frac{\partial}{\partial V}n_b^{rp} \tag{3.40}$$

4 Boundary Conditions

num.	name	variable	highest order derivative	center	edge
1	LQm1	Φ	2	N	0
2	LQm2	$\dot{\psi}_t$	0	×	×
3	LQm3	$\dot{\psi}$	0	×	×
4	LQm4	ψ	2	N	$2\pi\mu_0 I_\mathrm{p}$
5	LQm5	ψ_t/μ_0	2	0	$\langle\lvert\nabla V\rvert^2/R^2\rangle\lvert_{V_b} I_\mathrm{vac}/(4\pi^2)$
6	LQe1	n_e	2	N	N
7	LQe2	$n_\mathrm{e}u_\mathrm{e}^V$	1	0	×
8	LQe3	$\langle Bu_{\mathrm{e}\parallel}\rangle$	2(0)	N(×)	N(×)
9	LQe4	$n_\mathrm{e}\langle Ru_{\mathrm{e}\zeta}\rangle$	2	N	N
10	LQe5	$n_\mathrm{e}T_\mathrm{e}$	2	N	N
11	LQe6	$\langle B\hat{q}_{\mathrm{e}\parallel}\rangle$	0	×	×
12	LQe7	$n_\mathrm{e}\langle u_{\mathrm{e}\zeta}/R\rangle$	0	×	×
13	LQe8	$BV_{1\mathrm{e}}$	0	×	×
14	LQi1	n_i	2	N	N
15	LQi2	$n_\mathrm{i}u_\mathrm{i}^V$	1	0	×
16	LQi3	$\langle Bu_{\mathrm{i}\parallel}\rangle$	2(0)	N(×)	N(×)
17	LQi4	$n_\mathrm{i}\langle Ru_{\mathrm{i}\zeta}\rangle$	2	N	N
18	LQi5	$n_\mathrm{i}T_\mathrm{i}$	2	N	N
19	LQi6	$\langle B\hat{q}_{\mathrm{i}\parallel}\rangle$	0	×	×
20	LQi7	$n_\mathrm{i}\langle u_{\mathrm{i}\zeta}/R\rangle$	0	×	×
21	LQi8	$BV_{1\mathrm{i}}$	0	×	×
22	LQz1	n_z	2	N	N
23	LQz2	$n_\mathrm{z}u_\mathrm{z}^V$	1	0	×
24	LQz3	$\langle Bu_{\mathrm{z}\parallel}\rangle$	2(0)	N(×)	N(×)
25	LQz4	$n_\mathrm{z}\langle Ru_{\mathrm{z}\zeta}\rangle$	2	N	N
26	LQz5	$n_\mathrm{z}T_\mathrm{z}$	2	N	N
27	LQz6	$\langle B\hat{q}_{\mathrm{z}\parallel}\rangle$	0	×	×
28	LQz7	$n_\mathrm{z}\langle u_{\mathrm{z}\zeta}/R\rangle$	0	×	×
29	LQz8	$BV_{1\mathrm{z}}$	0	×	×
30	LQb1	n_b	0(2)	×(N)	×(N)
31	LQb2	$n_\mathrm{b}u_\mathrm{b}^V$	1	0	×
32	LQb3	$n_\mathrm{b}\langle Bu_{\mathrm{b}\parallel}\rangle$	0	×	×
33	LQb4	$n_\mathrm{b}\langle Ru_{\mathrm{b}\zeta}\rangle$	0	×	×
34	LQb7	$n_\mathrm{b}\langle u_{\mathrm{b}\zeta}/R\rangle$	0	×	×
35	LQb8	$BV_{1\mathrm{b}}$	0	×	×
36	LQn1	n_{01}	2	N	$\langle\lvert\nabla V\rvert\rangle S_\mathrm{gas}\rvert_b$
37	LQn2	n_{02}	2	N	0
38	LQn3	n_{03}	2	N	0
39	LQnz	$\bar{n}_z$	2	N	$\langle\lvert\nabla V\rvert\rangle S_{\mathrm{gas,z}}\rvert_b$
40	LQr1	n_b^rp	0(2)	×(N)	×(N)

0: Dirichlet condition ($u = 0$), N: Neumann condition ($u' = 0$)
value: Neumann condition (u' = value), ×: no condition

NOTE: The dependent variable of LQm4 turns to be ψ/μ_0 when NB is applied. Subsequently, the edge boundary condition listed for that variable is also changed to $2\pi I_\mathrm{p}$.

NOTE: Rows 2 to 5 deserve a word, because the variable that carries the second-order operator is not the variable of the equation in which the operator is written. The LQm2 equation advances $\dot{\psi}_t$ and contains $\frac{\partial}{\partial V}[\langle|\nabla V|^2/R^2\rangle \langle R^{-2}\rangle^{-1} \frac{\partial}{\partial V}(\psi_t/\mu_0)]$; the LQm3 equation advances $\dot{\psi}$ and contains $\frac{\partial}{\partial V}[\langle|\nabla V|^2/R^2\rangle \frac{\partial}{\partial V}\psi]$. Counting per dependent variable over the whole system, as this section does, $\dot{\psi}_t$ and $\dot{\psi}$ are never differentiated and so take order 0 and no condition of their own, while ψ_t/μ_0 and ψ take order 2 and carry the two conditions each. The table above now reads that way. In the implementation the two conditions of each pair are applied to the LQm2 and LQm3 equation rows, since that is where the boundary term left by the integration by parts appears; this is the correct weak form and is not a discrepancy with the table. The axis condition of ψ_t/μ_0 is imposed there as $\dot{\psi}_t|_{V=0} = 0$, which holds $\psi_t(0)$ at its initial value, namely zero, and is therefore the Dirichlet condition listed.

NOTE: In the table the column labeled by the highest order derivative means the order of the highest order derivative of a dependent variable, not an equation, in the corresponding row. Please note that the term not integrated by parts is never counted. For example, the first row shows the highest order of derivatives is equal to 2 for Φ because the LQm1 equation has the diffusion term having the second derivative of Φ with respect to V. In addition, the LQe2 and LQi2 equations also have the first derivative of Φ, which is not a dependent variable of these equations. As a whole, the highest order derivative of Φ in this equation system is 2 that comes from the LQm1 equation. When we look at n_e, we find that the highest order derivative of n_e is 2 shown in the LQe5 equation, even though no derivative appears in the LQe1 equation. The Neumann conditions for n_e come from the LQe5 equation. When it comes to $n_\mathrm{e}u_\mathrm{e}^V$, the highest derivative is found as order 1 in the LQe1 equation, but this term is not integrated by parts. It means that any boundary condition is not naturally imposed on $n_\mathrm{e}u_\mathrm{e}^V$. The Dirichlet boundary condition should be imposed at axis.